VLA / EMBODIED AI

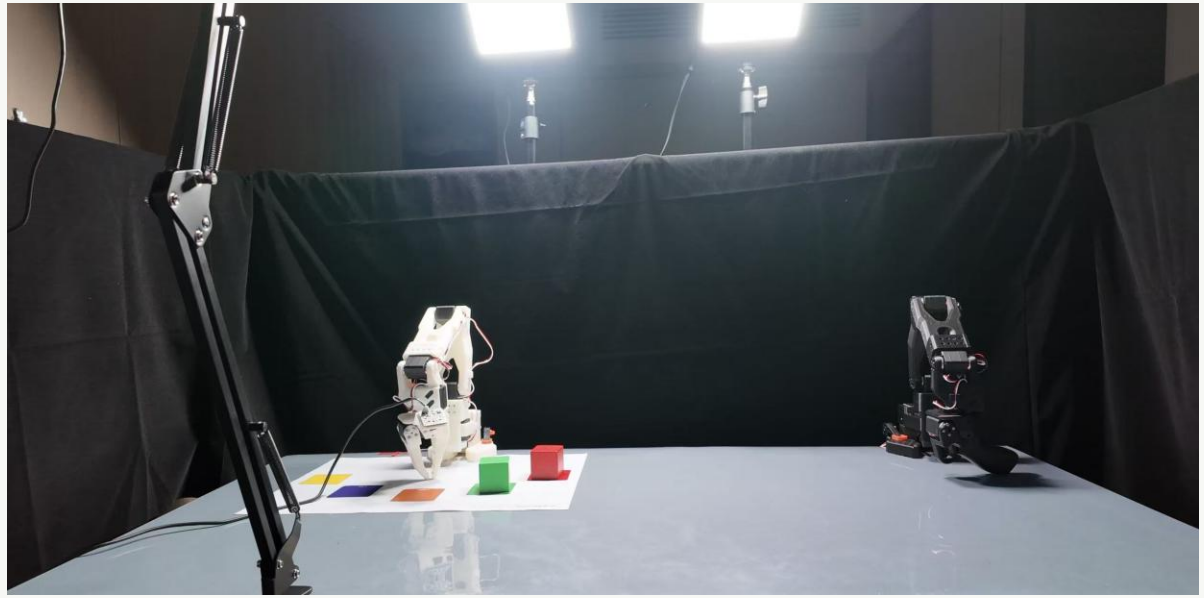
Reliable embodied action starts with evidence

Multimodal diagnosis, real-robot calibration and closed-loop control

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BEng Mechanical Engineering, UCL
2024–2027

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Contents

Projects connect evidence diagnosis to embodied action

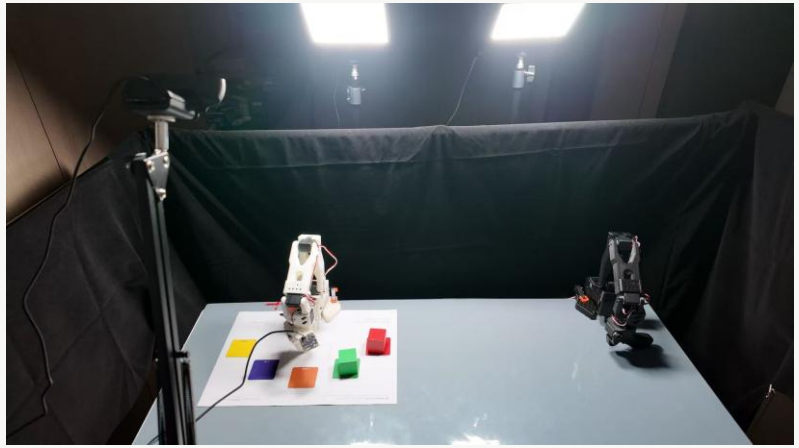
	PROJECT	RESEARCH FOCUS	PAGES
01	**Fudan VLA**	Policy reliability under shift	06–10
02	**PercepAgent**	Tool selection from evidence gaps	11–14
03	**Fudan calibration**	Pose correction before contact	15–17
04	**RSG-VLM**	Metric reasoning with radar graphs	18–24
05	**Navigation and recovery**	Explicit control mode switching	25–26
06	**Research plan**	Residual evidence and recovery	27–30

Overview 03–05 Closing 31 Appendix 32–39

Research profile

UCL Mechanical Engineering, 2024–2027 / robotics and multimodal reasoning

71.84%	ACM MM	3 papers	3 settings	1 showcase
UCL average First Class expected	2026 accepted First author / CCF-A	1 published 2 accepted	Fudan / Cambridge UCL	Humanoid robotics CISMA



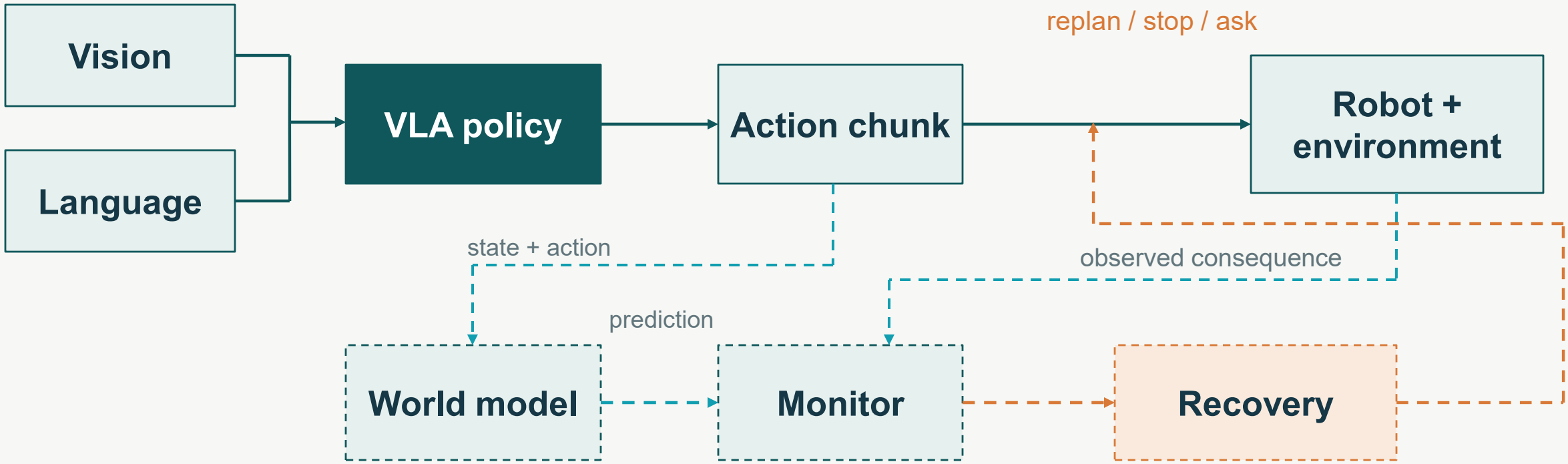
Research path

2024	UCL BEng
2025	Fudan calibration / first publication
2026	VLA experiments / ACM MM accepted
2027	BEng completion (expected)

Evidence diagnosis and robot deployment define the research direction

Reliable VLA needs a monitor around the action loop

AutoMan fit: VLM/VLA, world models and human oversight / proposed integration



Dashed paths: proposed assurance layer

The monitor tests whether observed consequences still support execution

RESEARCH PROJECT

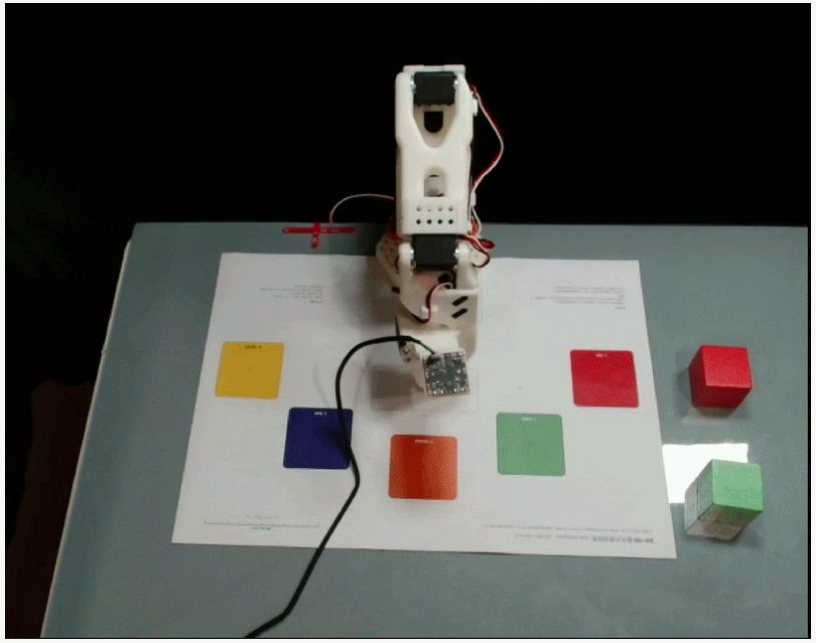
01 Fudan VLA

Real-robot policy evaluation under distribution shift

Which failures should drive the next demonstration set?

A repeatable experiment loop makes VLA testable

Fudan robotics lab / SO-ARM101 / ACT baseline and SmolVLA / LeRobot + PyTorch



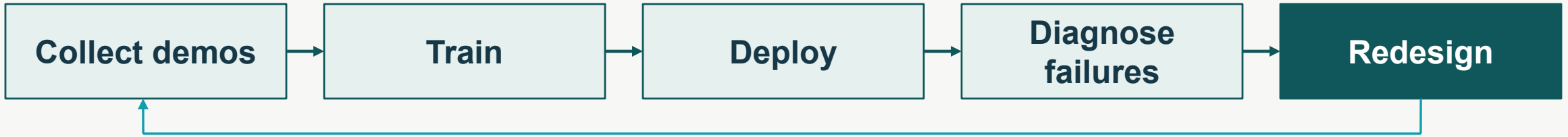
≈280 → **240**

raw demonstrations — curated episodes

3 tasks

2 RGB views / 30 Hz

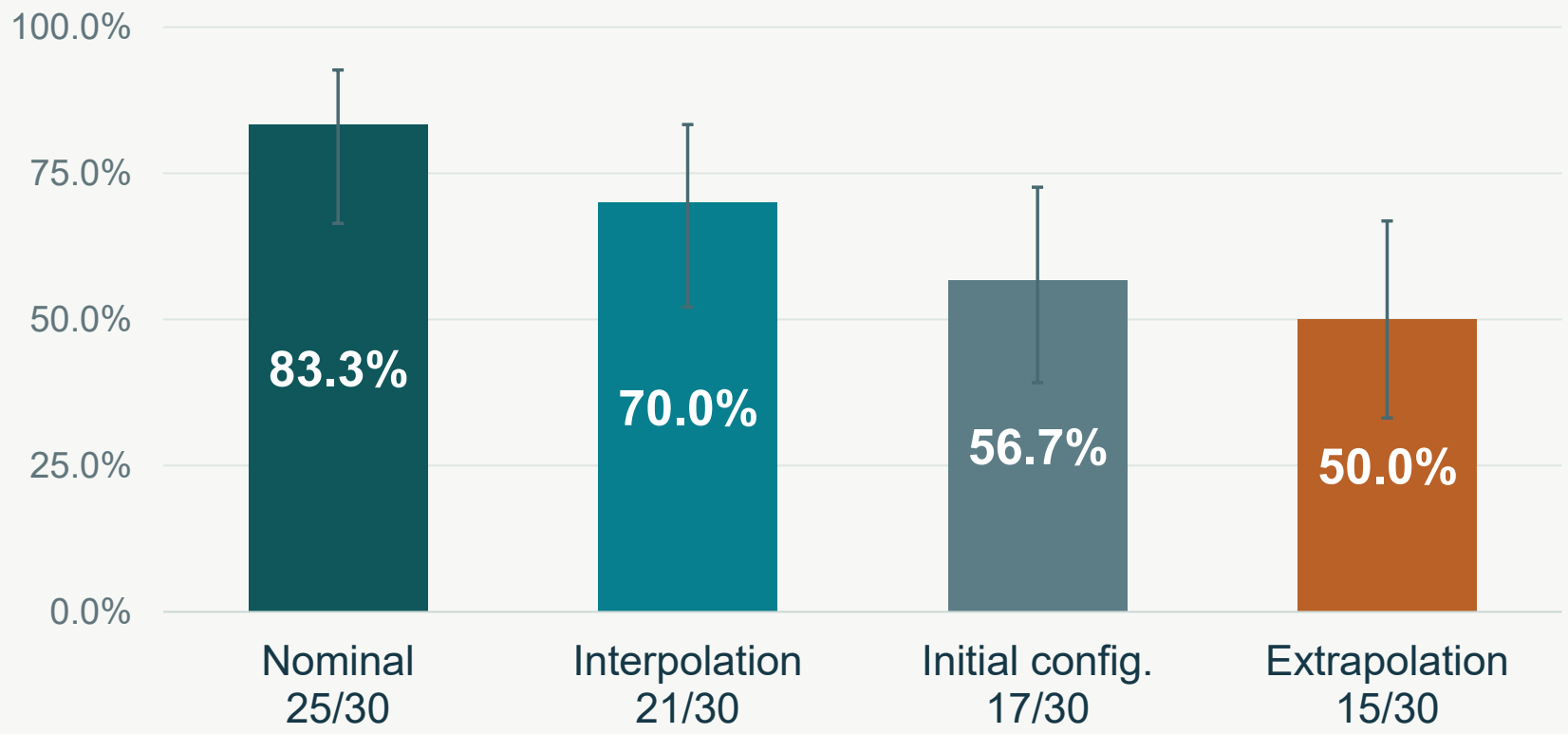
My role: experiment design and iteration



Controlled evaluation determines what the next collection round must change

Reliability degrades beyond demonstration coverage

SmolVLA / T1 + T2 pooled / 120 pre-specified trials / 30 per condition



83.3%

nominal success

50.0%

extrapolation success

p ≈ 0.006

pose-shift trend

Error bars: Wilson 95% confidence intervals

The experiment supports degradation, with overlapping intervals under shift

70% of shifted failures occur before contact

First observable failure event / 42 failed episodes in 120 trials

	PRE-CONTACT					
Condition	**Approach**	**Alignment**	**Grasp**	**Transport**	**Temporal**	**Total**
Nominal	**0**	**2**	**2**	**1**	**0**	5
Interpolation	**2**	**4**	**2**	**1**	**0**	9
Initial config.	**5**	**4**	**1**	**1**	**2**	13
Extrapolation	**6**	**5**	**2**	**1**	**1**	15
Total	**13**	**15**	**7**	**4**	**3**	**42**

26 / 37

shifted failures
at approach or alignment

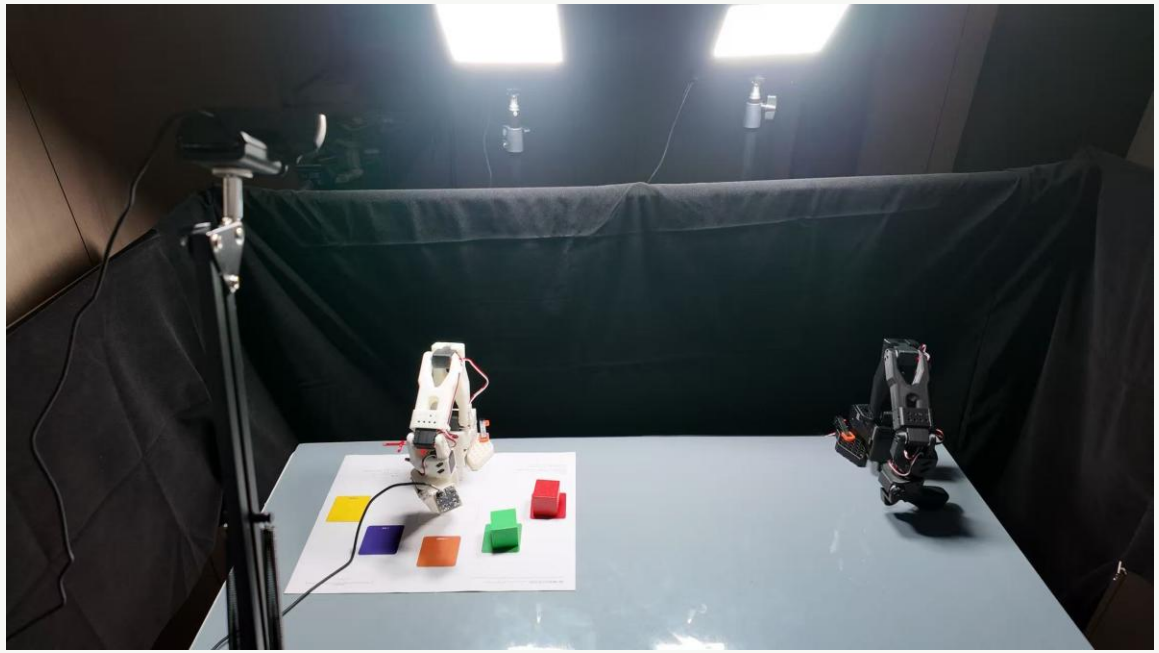
88%

initial label agreement
annotator + full reviewer

Locus labels guide interventions; causal attribution needs additional controls

Pose shifts reveal imprecise spatial adaptation

24 failed episodes with shifted object poses / endpoint comparison

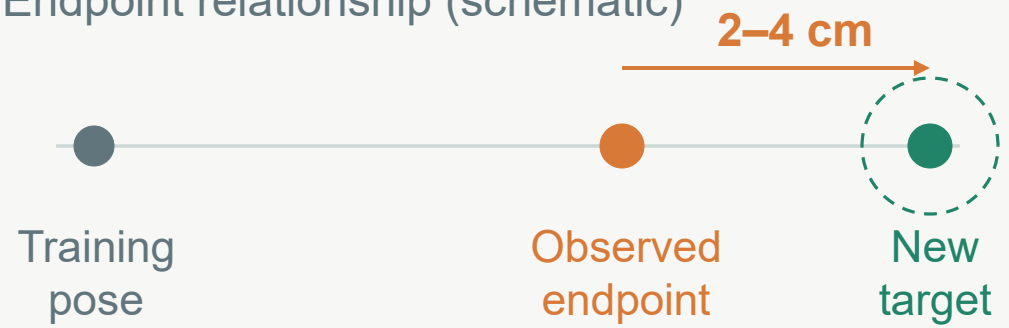


SO-ARM101 workspace

17 / 24

endpoints closer to the new target
than the nearest training pose

Endpoint relationship (schematic)



<1.5 cm grasp tolerance

The policy moves toward shifted targets, but often misses grasp tolerance

02 PercepAgent

Evidence diagnosis and selective visual tools

Can the right evidence recover a frozen model’s answer?

The right tool exposes evidence the baseline misses

PercepAgent / ACM MM 2026 accepted / first author / four observed evidence gaps

Resolution



Super-resolution



Depth



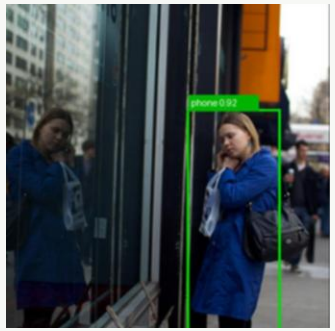
Depth map



Localization



Grounding



Region semantics



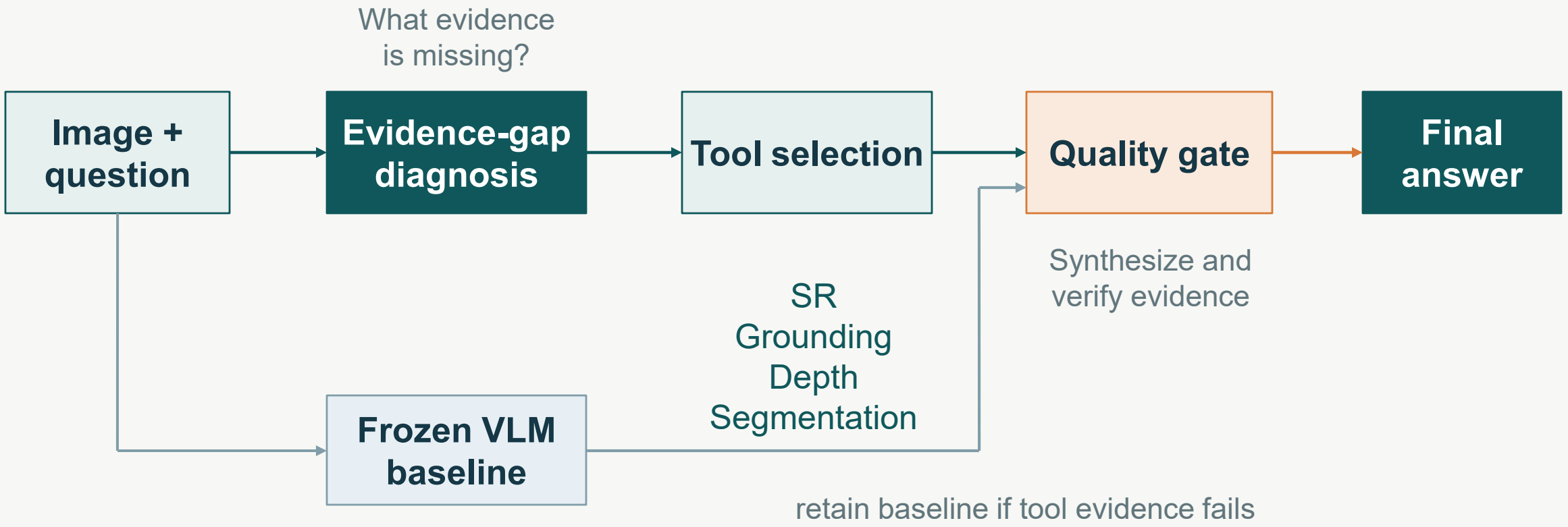
Segmentation



Different evidence gaps require different perceptual interventions

Evidence diagnosis drives selective tool use

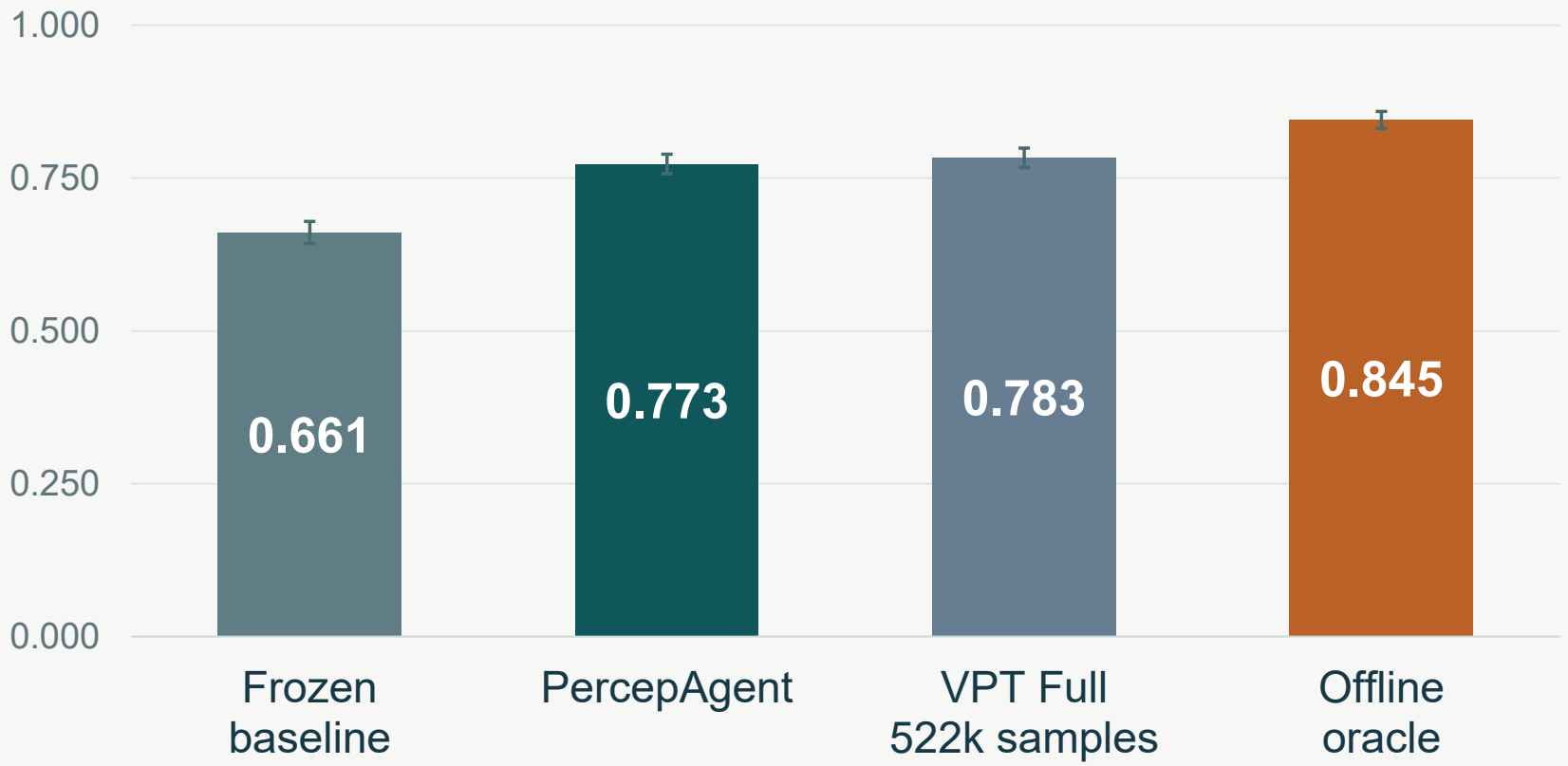
PercepAgent / frozen backbone / no additional training



Tool choice follows the missing evidence, with an explicit fallback path

PercepAgent recovers 61% of the oracle headroom

Qwen2-VL-7B / average GPT-4o semantic score across five benchmarks



+16.9%

relative gain over baseline

61%

headroom recovered

0.806

Qwen3: +10.7%
0.728 baseline
no reconfiguration

0.112 gain from 0.184 measured headroom, with a frozen backbone

03 Fudan calibration

External pose correction for mobile manipulation

How can the robot correct its grasp after docking?

External calibration makes the grasp pose observable

Fudan manufacturing robot / external virtual TCP and coordinate registration

Physical constraint



Dock error >1 mm / >1°

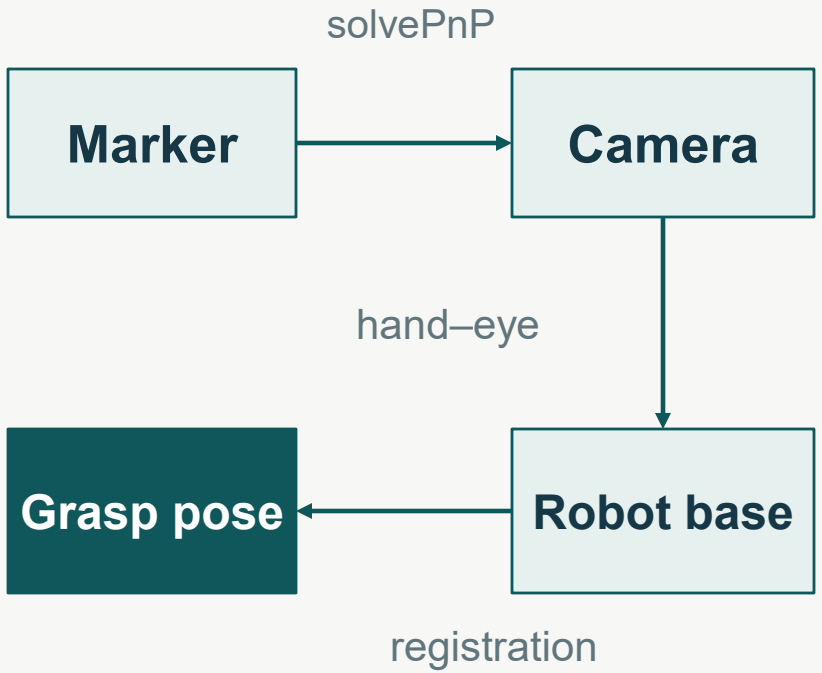
Blocked calibration route

TCP calibration

Unavailable

Precise contact still requires a pose reference

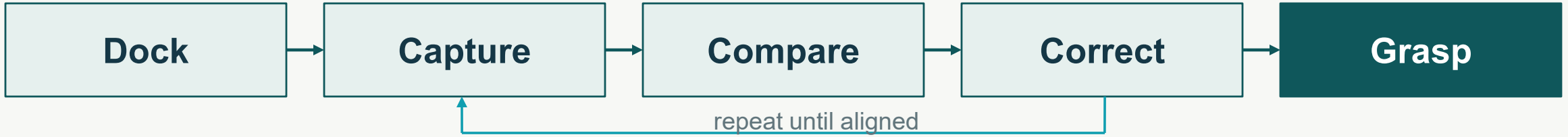
External pose chain



An observable coordinate chain enables correction before physical contact

Pose correction closes the docking-to-grasp loop

Fudan mobile manipulation / ROS + MoveIt integration / CISMA exhibition









±0.5 mm
grasping accuracy

<0.5°
rotation error
after correction

ROS / MoveIt
perception and
grasping integration

CISMA
system exhibited

Perception becomes useful when it changes the executed grasp

04 RSG-VLM

Radar scene graphs for auditable metric reasoning

Which spatial claims survive changes in the input interface?

Radar geometry becomes an auditable model input

RSG-VLM / co-first author / ECCV 2026 DriveX Workshop / interface study

A Image only

RGB image

No metric sensor

B Radar caption

Free-form description

Same six targets

C Category graph



front-right
medium
receding

supplied output labels

D Numeric graph



19.0 m
+29°
+5.66 m/s

measurements only

One output schema: direction / range / radial state / hazard cue

Units and provenance make metric statements traceable to measurements

Numeric graphs expose the measurements behind each cl

Same six targets / three shown / +y forward, +x right / radial velocity <0 means closing

Numeric graph: measurements with explicit units

Target	Forward (m)	Right (m)	Range (m)	Azimuth (°)	Radial (m/s)
t21 / car	**16.6**	**+9.2**	**19.0**	**+29**	**+5.66**
t93 / car	21.4	−4.3	21.9	−11	+3.16
t43 / pedestrian	3.3	+22.0	22.2	+81	−0.12

Category graph / t21

front-right / medium / receding / low

Scene cue: medium

Shared output vocabulary

SCENE	DIRECTION	RANGE	MOTION	RISK
one sentence	6 zones	3 bins	5 states	3 levels

The category graph supplies labels; the numeric graph supplies measurements

Structured radar improves metric-field accuracy

404 nuScenes-mini keyframes / independent reference / same six targets

Condition	Direction	Range	Radial	Hazard
Rule only	**1.00**	**1.00**	**0.57**	**0.66**
GPT / image	0.18	0.74	0.37	0.32
GPT / caption	1.00	0.85	0.57	0.51
GPT / category	**1.00**	**1.00**	**0.57**	**0.66**
Qwen / image	0.18	0.71	0.38	0.41
Qwen / caption	0.76	0.77	0.55	0.37
Qwen / category	**0.98**	**1.00**	**0.58**	**0.67**

+0.23

Qwen range gain
graph versus caption

$p < 10^{-4}$

range, graph vs caption
paired McNemar tests

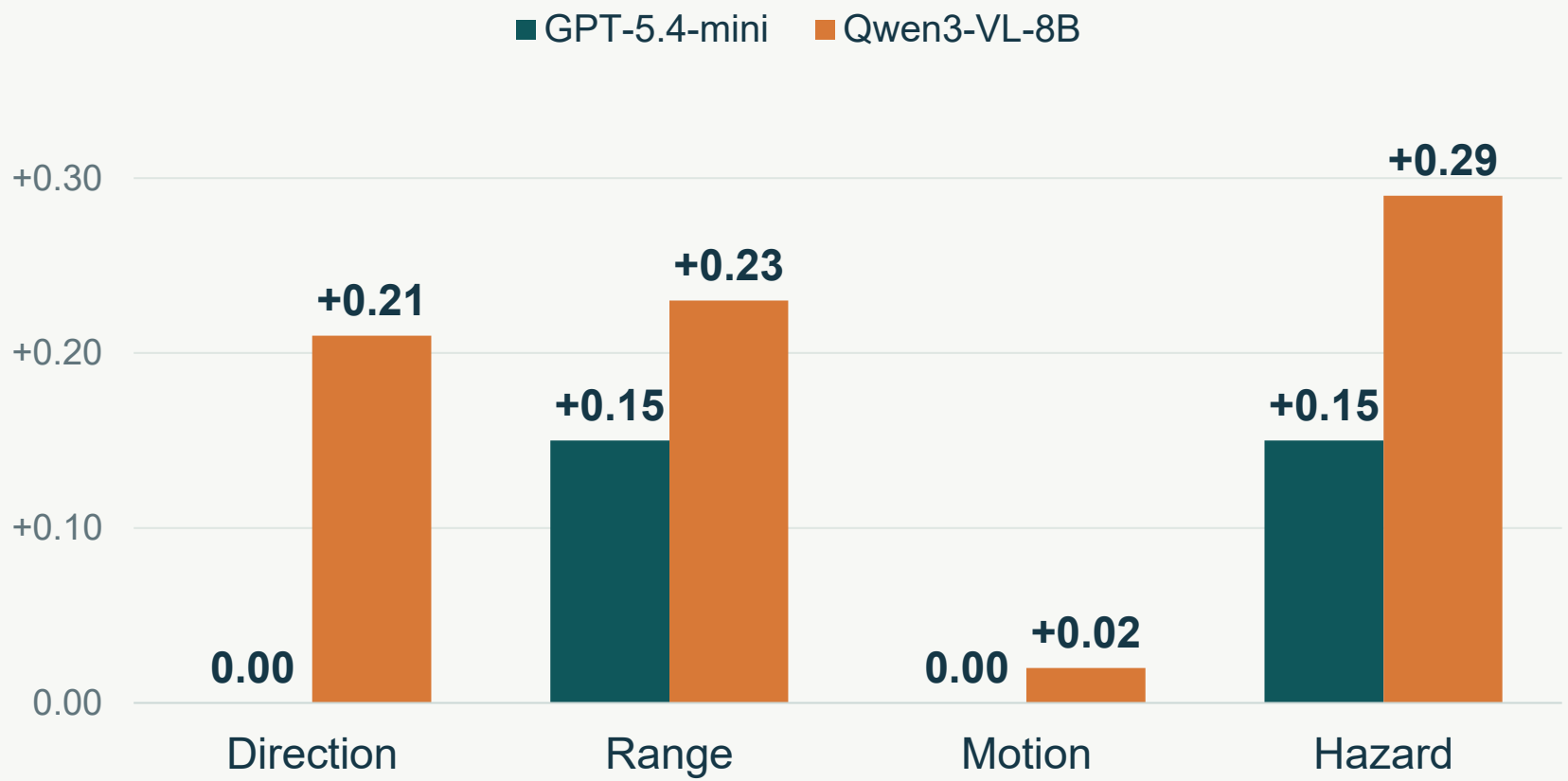
0.56 ms

construction per frame

Category graphs approach rule-only accuracy by supplying auditable labels

Radar graphs improve range and hazard cues

404-keyframe study / category graph minus radar caption / source-reported gains



+0.23

Qwen range accuracy

+0.29

Qwen hazard-cue accuracy

+0.15

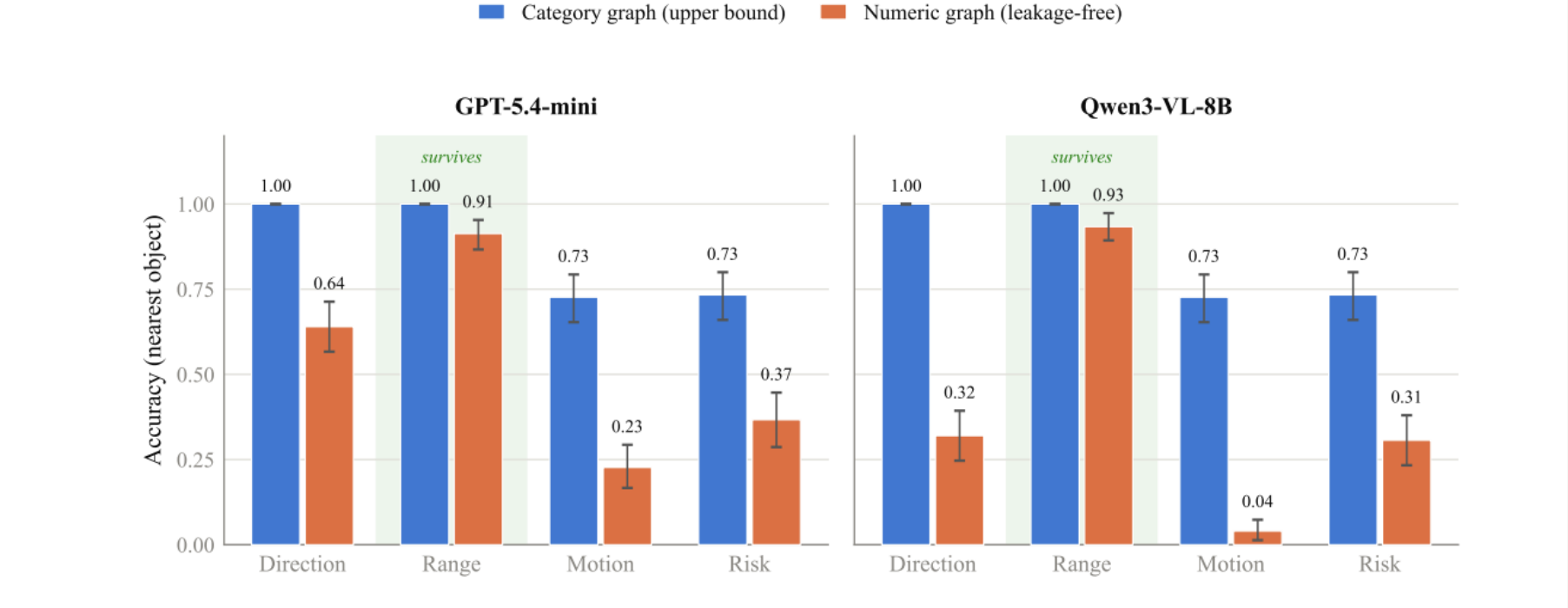
GPT range and hazard-cue accuracy

Figure-reported deltas; rounded-table subtraction differs by ≤0.01

Structured radar evidence adds more than a free-form radar description

Range survives when category labels are removed

150 matched keyframes / numeric-only graph / category graph as transcription upper bound



Original source figure (non-editable image); editable values in appendix

0.91 / 0.93

range accuracy survives without supplied labels

0.04

Qwen radial-state accuracy

≤0.04

change without image

Range remains usable; direction and motion need additional grounding

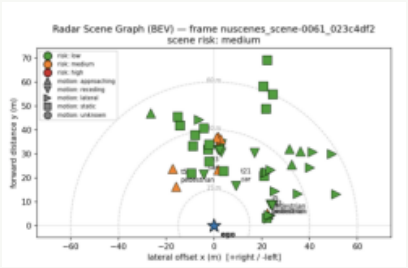
Radar grounding makes scene-level errors visible

scene-0061 / nearest-object comparison / hazard is a surrogate cue

Camera frame



Radar BEV



Condition	Direction	Range	Motion	Hazard
Reference	front-right	medium	receding	low
Image only	front-left	near	static	high
Radar caption	front-right	near	receding	low
Category graph	front-right	medium	receding	low

0 / 4
Image only

3 / 4
Radar caption

4 / 4
Category graph

The scene comparison ties each corrected field to measured radar evidence

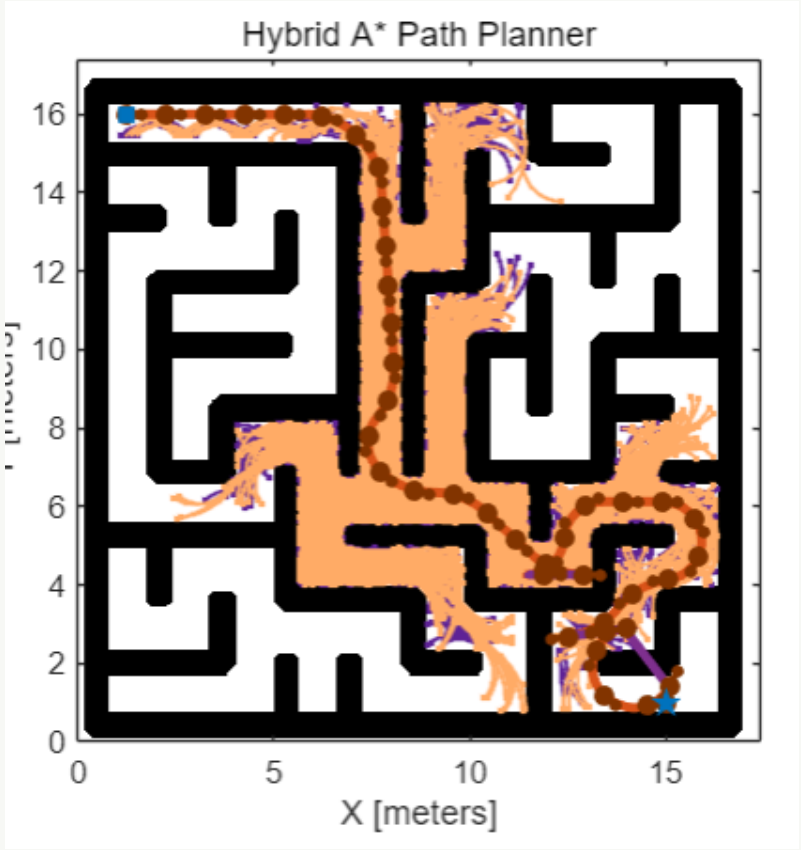
05 Navigation and recovery

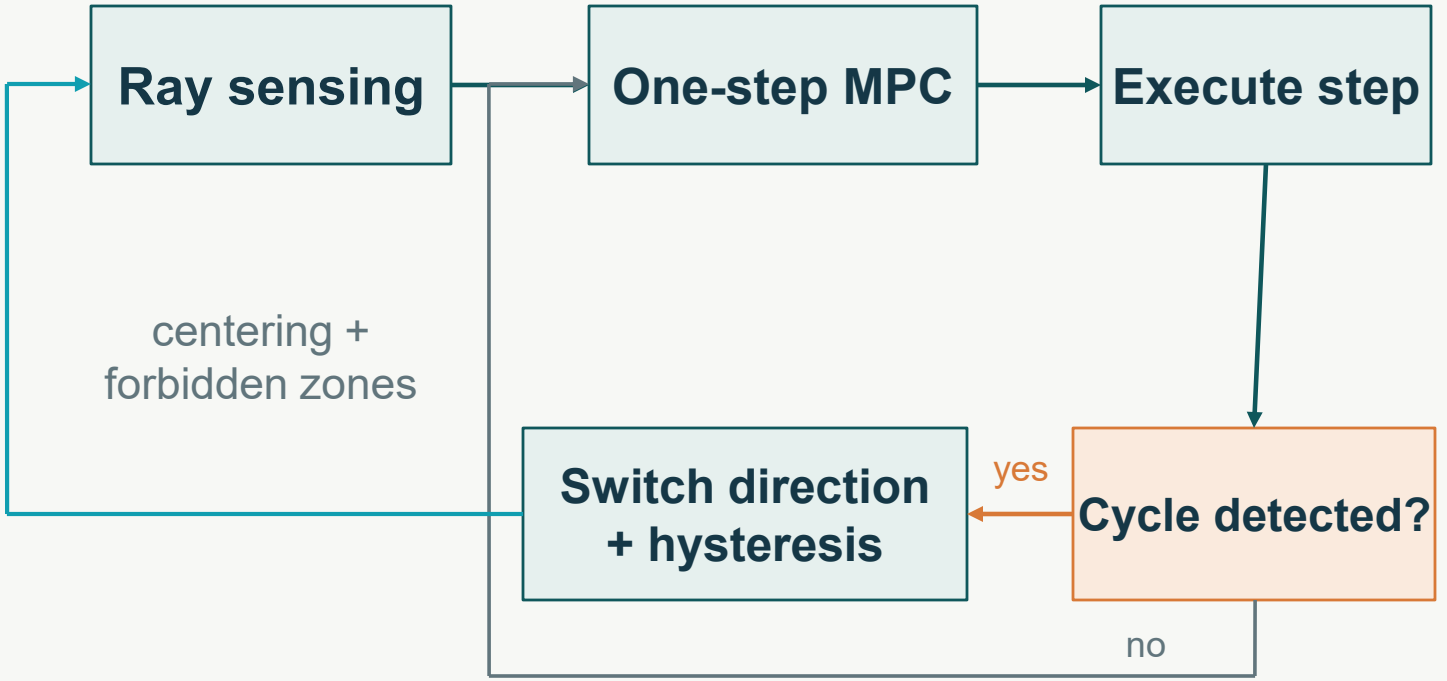
Control decisions in a maze navigation simulation

When should execution switch to a recovery mode?

Recovery is an explicit control decision

Cambridge-guided project / navigation simulation / multi-sensor fusion review





52-paper VLA survey

First-author fusion review

Cycle detection triggers direction changes and local centering corrections

06 Research plan

World-model residuals, validity boundaries and recovery

When does a residual become useful evidence for action?

Residual trust requires three linked empirical tests

Proposed research / incremental evidence, statistical validity and closed-loop utility

WP1
Matched evidence

→

WP2
Validity boundary

→

WP3
Transfer + recovery

Novelty × failure controls

	No failure	Failure
Familiar	control	injected
Novel	control	injected

$$B_0 \subset B_1 \subset B_2$$

Observed − predicted

Estimation uncertainty

Calibrated evidence

6 closed-loop arms

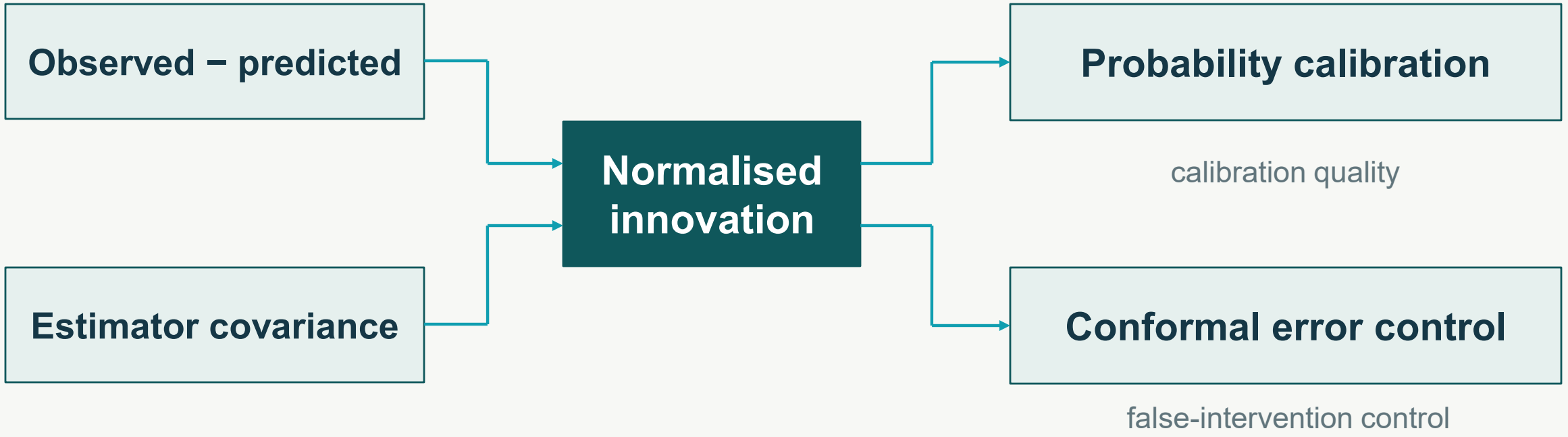
Oracle-onset control

Source-fixed alarm rule
Franka-class confirmation

Each package tests a necessary link between residuals and useful recovery

A large residual is insufficient failure evidence

Proposed WP2 / validity axes: estimation uncertainty, novelty and contact



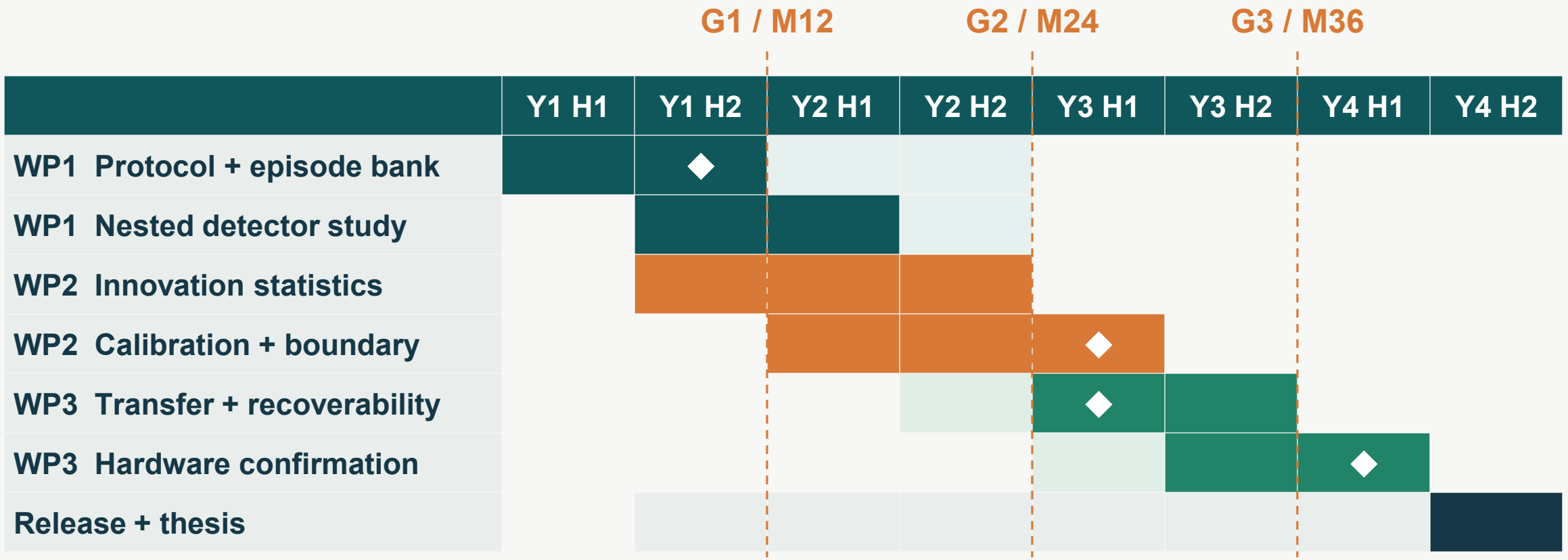
Pre-registered boundary

Supported	Non-useful	Indeterminate

The output is a validity boundary, conditional on the operating regime

Four-year gates preserve useful research outcomes

Proposed roadmap / half-year resolution / G1 M12, G2 M24 and G3 M36



◆ paper

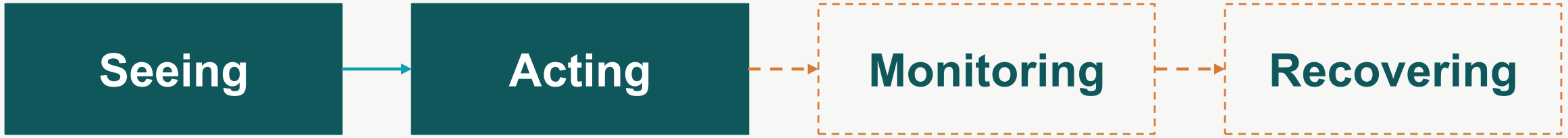
Lighter fill: preparation and parallel work

Matched null results still yield a protocol, boundary map and transfer study

Reliable VLA needs evidence throughout execution

A research path connecting completed studies with a proposed assurance programme

COMPLETED STUDIES **PROPOSED RESEARCH**



Evidence diagnosis
PercepAgent, RSG-VLM

Calibration
Real-robot deployment

Validity boundary
WP2

Transfer + fallback
WP3

61%
headroom recovered

±0.5 mm
reported grasp accuracy

When should the robot trust its evidence?

Thank you
CLOSING

shangze.kong.24@ucl.ac.uk

A Additional evidence

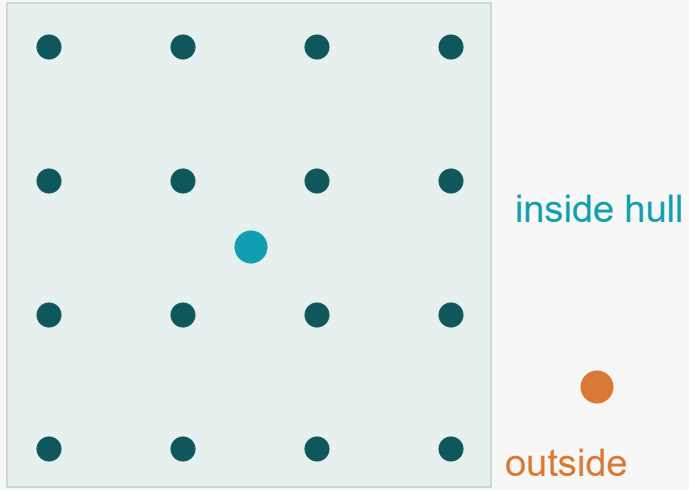
Protocols, ablations and qualitative comparisons

Supporting material for discussion

The evaluation separates selection from reporting

Appendix / training coverage, held-out shifts and a language control

Training coverage



4 × 4 grid / 4 yaw orientations

≈6.5 cm grid spacing
2–3 demos per pose–yaw

Protocol schematic, not trial coordinates

Condition	Shift	Trials
Nominal	evaluation reference	30
Interpolation	3–5 cm inside hull	30
Initial config.	joints 10–20° EE start 8–15 cm	30
Extrapolation	planar 5–10 cm depth 8–12 cm	30

7 / 10

target-switch instructions followed (limited probe)



Diagnosis and verification both contribute to the gain

Appendix / PercepAgent ablations and pairwise tool complementarity

Component ablation

Variant	Score	Δ vs full
Full selector	0.773	—
No diagnosis	0.691	−10.6%
No commit gates	0.733	−5.2%
No verification	0.752	−2.7%

Tool complementarity (%)

	Base	SR	GD	Depth	Seg
Base	—	15.4	11.6	9.3	10.9
SR	15.4	—	17.0	13.1	14.3
GD	11.6	17.0	—	13.0	14.4
Depth	9.3	13.1	13.0	—	7.7
Seg	10.9	14.3	14.4	7.7	—

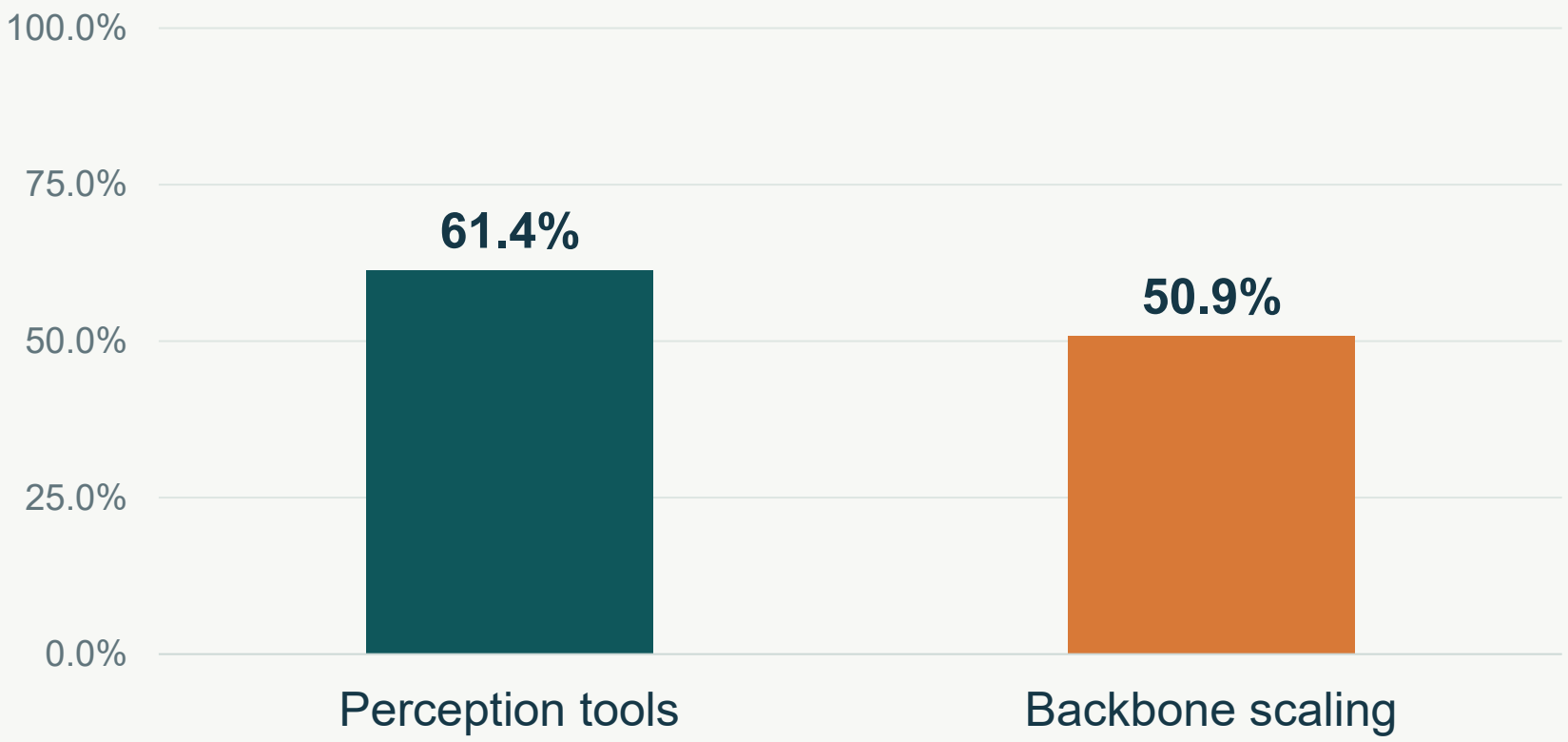
Remaining failure split

45% integration **39%** routing **16%** tool mismatch

Correct tool selection still requires faithful integration of the evidence

Tools recover more baseline errors than model scaling

Appendix / aggregate failure attribution / tools versus Qwen3 backbone scaling



10.5 pp

more baseline errors fixed

77.1%

capacity-failure subset fixable by the oracle

+0.001

mean-score gain from confidence routing

Failure attribution changes when information access becomes a measured control

Full benchmark comparison across both backbones

Appendix / Q2 = Qwen2-VL-7B / Q3 = Qwen3-VL-8B / scores with source confidence intervals

Method	Train	Tools	CUB	Flickr	GQA	POPE	VSR	Avg.	Δ rel.
Q2 / Baseline	none	0	0.558 ±0.043	0.634 ±0.032	0.608 ±0.042	0.867 ±0.033	0.640 ±0.043	0.661 ±0.018	—
Q2 / VPT (GD)	522k	1	0.930 ±0.022	0.720 ±0.030	0.590 ±0.042	0.883 ±0.031	0.730 ±0.040	0.771 ±0.016	+16.6%
Q2 / VPT (Full)	522k	2	0.921 ±0.024	0.680 ±0.031	0.623 ±0.042	0.903 ±0.029	0.788 ±0.036	0.783 ±0.016	+18.5%
Q2 / PercepAgent	**none**	**4**	**0.800 ±0.035**	**0.740 ±0.029**	**0.673 ±0.041**	**0.940 ±0.023**	**0.710 ±0.040**	**0.773 ±0.016**	**+16.9%**
Q2 / Offline oracle	none	4	0.919 ±0.024	0.790 ±0.027	0.753 ±0.037	0.945 ±0.022	0.817 ±0.034	0.845 ±0.014	—
Q3 / Baseline	none	0	0.720 ±0.039	0.814 ±0.026	0.520 ±0.043	0.900 ±0.029	0.687 ±0.041	0.728 ±0.017	—
Q3 / PercepAgent	**none**	**4**	**0.850 ±0.031**	**0.820 ±0.026**	**0.710 ±0.039**	**0.950 ±0.021**	**0.700 ±0.041**	**0.806 ±0.015**	**+10.7%**
Q3 / Offline oracle	none	4	0.940 ±0.021	0.850 ±0.024	0.743 ±0.038	0.940 ±0.023	0.852 ±0.032	0.865 ±0.013	—

PercepAgent improves both backbones without additional training

Evidence diagnosis outperforms the routing controls

Appendix / Qwen2-VL-7B / no additional training / source benchmark means and intervals

Strategy	Tools	CUB	Flickr	GQA	POPE	VSR	Avg.	Δ rel.
SR only	1	0.678 ±0.041	0.694 ±0.031	0.635 ±0.042	0.905 ±0.028	0.672 ±0.042	0.717 ±0.017	+8.5%
Grounding only	1	0.597 ±0.043	0.565 ±0.033	0.618 ±0.042	0.919 ±0.026	0.648 ±0.043	0.669 ±0.018	+1.2%
Depth only	1	0.697 ±0.040	0.658 ±0.032	0.624 ±0.042	0.913 ±0.027	0.691 ±0.041	0.717 ±0.017	+8.3%
Segmentation only	1	0.756 ±0.037	0.649 ±0.032	0.651 ±0.041	0.916 ±0.027	0.680 ±0.042	0.731 ±0.017	+10.6%
Random-tool	4	0.740 ±0.038	0.698 ±0.031	0.560 ±0.043	0.900 ±0.029	0.676 ±0.042	0.715 ±0.017	+8.2%
All-tools	4	0.760 ±0.037	0.750 ±0.029	0.600 ±0.042	0.910 ±0.028	0.592 ±0.044	0.722 ±0.017	+9.2%
ReActAgent	4	0.690 ±0.040	0.591 ±0.032	0.610 ±0.041	0.840 ±0.035	0.580 ±0.044	0.662 ±0.018	+0.1%

PercepAgent 0.773

Best single tool 0.731

Tool access helps; evidence diagnosis improves which evidence gets used

Five benchmarks probe different visual evidence gaps

Appendix / Qwen2-VL-7B / GPT-4o semantic scores on [0,1] / bootstrap evaluation

Benchmark	n	Task / evidence	Baseline	Agent	Oracle
CUB-200	492	Bird attributes resolution	0.558	**0.800**	0.919
Flickr30K	1,546	Appearance + clothing local detail	0.634	**0.740**	0.790
GQA	500	Compositional reasoning mixed deficits	0.608	**0.673**	0.753
POPE	2,910	Object hallucination localization	0.867	**0.940**	0.945
VSR	404	Spatial relationships depth / geometry	0.640	**0.710**	0.817

A shared protocol spans fine detail, object hallucination and geometry

Range survives when category labels are removed

150 matched keyframes / numeric-only graph / category graph as transcription upper bound

Condition	Direction	Range	Radial	Hazard
GPT / category	1.00	1.00	0.73	0.73
GPT / numeric	**0.64**	**0.91**	**0.23**	**0.37**
Qwen / category	1.00	1.00	0.73	0.73
Qwen / numeric	**0.32**	**0.93**	**0.04**	**0.31**

GPT-5.4-mini and Qwen3-VL-8B

0.91 / 0.93

range accuracy survives without supplied labels

0.04

Qwen radial-state accuracy

≤0.04

change without image

Reliable interfaces still need deterministic measurement-to-symbol conversion